

BCCC CSV merge
(fraud + legitimate
DeFiTransLyzer)

preprocess.py
21 behavioral features;
drop flag;
group split by address
(seed 42)

Train: RandomForest (80
trees)
+ IsolationForest + Logistic
stacking

Artifacts: model .pkl,
metrics.json,
shap_global.json,
confusion_matrix

Export figures:
metrics-bars, confusion-
matrix, shap-importance

Thesis Figure ml-eval
(BCCC subsample;
synthetic: false)